SOURAJIT MAJUMDER

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EDUCATION

Narula Institute of Technology (2023 – 2027)
B.Tech in Computer Science Engineering (AI & ML) | CGPA: 8.51/10.0
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Artificial Intelligence, Database Management Systems

St. Augustine's Day School, Barrackpore (ISC) (2023)
Higher Secondary (Grade 12): 94%

St. Augustine's Day School, Barrackpore (ISC) (2021)
Secondary (Grade 10): 90%

CERTIFICATIONS

- Certified System Administrator (CSA) – ServiceNow (Issued: June 2026)

PROJECTS

Glaucoma Detection System — Deep Learning, Computer Vision (2026)
github.com/Sourajit-M/glaucoma-detection-project | Live Demo

- Built AI based Glaucoma detection system using ResNet-18 and U-Net with an AUC of 94.5% and accuracy of 87.1% which outperforming the strongest classical baseline (SVM, 78.9% AUC) by 15.6 percentage points.
- Designed end-to-end ML pipeline comprising of image pre-processing, data augmentation, model training, Grad-CAM explainability and deployment as a real-time inference API.
- Technologies Used: Python, PyTorch, ResNet-18, U-Net, Grad-CAM, scikit-learn, OpenCV

YouTube Semantic Intelligence & Search Engine (2025)
github.com/Sourajit-M/youtube-semantic-search | Live Demo

- Built an automated pipeline for ingestion, chunking and indexing of YouTube video transcripts, to support a grounded Q&A; engine that answers with exact source citations.
- Implemented a hybrid retrieval system (BM25 + semantic search with RRF reranking), reducing irrelevant/incorrect retrievals by approximately 40% versus keyword-only search.
- Technologies Used: Python, FastAPI, ChromaDB, LiteLLM, SQLite, React

Local AI Inference Platform — (FastAPI, React, Ollama) (2026)
github.com/Sourajit-M/local-ai-inference-platform | Live Demo

- Developed a local AI inference platform for fully offline SLM execution, with JWT authentication, persistent chat sessions, structured generation, and experiment tracking.
- Built a benchmarking framework evaluating Qwen 2.5 3B, Llama 3.2 3B, and Phi-3 Mini on consumer hardware (RTX 4050, 4GB VRAM); optimized champion model (Qwen 2.5 3B) achieved 0.81s TTFT and 40.17 tok/s throughput, visualized via a React analytics dashboard.
- Technologies Used: Python, FastAPI, TypeScript, Ollama, SQLite, Pydantic, JWT

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript
- **AI / ML & Computer Vision:** scikit-learn, LangChain, LangGraph, RAG, ChromaDB
- **Web & Backend:** FastAPI, Node.js, Express.js, REST API, MongoDB, PostgreSQL
- **Tools & Platforms:** Git, GitHub, Docker
- **Core CS:** Data Structures & Algorithms, OOP, System Design Fundamentals

ACHIEVEMENTS & EXTRACURRICULARS

- **LeetCode:** Solved 450+ problems spanning dynamic programming, graphs, trees, and binary search.
- **Certification:** Completed Full Stack Web Development (100xDevs) – React, Node.js, PostgreSQL, and system design fundamentals.

• **Hackathon:** Participated in Smart India Internal Hackathon (SIH) 2025 and worked with a team to design and build a working AI-based prototype within a fixed hackathon timeline.